SQL Coding Challenge:

College Student Management System

Created Tables:

Students Table, Courses Table, Marks Table

```sql
create database college;
use College;
CREATE TABLE students
(student_id INT PRIMARY KEY, student_name VARCHAR(50), gender VARCHAR(10), city VARCHAR(50), join_year INT);
CREATE TABLE courses
(course_id INT PRIMARY KEY, course_name VARCHAR(100), department VARCHAR(50));
CREATE TABLE marks
(mark_id INT PRIMARY KEY, student_id INT, course_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id));
describe students;
describe courses;
```

Inserted Values:

```sql
create database college;
use College;
CREATE TABLE students
(student_id INT PRIMARY KEY, student_name VARCHAR(50), gender VARCHAR(10), city VARCHAR(50), join_year INT);
CREATE TABLE courses
(course_id INT PRIMARY KEY, course_name VARCHAR(100), department VARCHAR(50));
CREATE TABLE marks
(mark_id INT PRIMARY KEY, student_id INT, course_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id));
describe students;
describe courses;
describe marks;
INSERT INTO students
(student_id, student_name, gender, city, join_year)
VALUES (1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023),
(3, 'Kiran', 'M', 'Tumakuru', 2024), (4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses
(course_id, course_name, department) VALUES (101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks
(mark_id, student_id, course_id, marks) VALUES (1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
```

1. Display all students

```
create database college;
use college;
CREATE TABLE students ( student_id INT PRIMARY KEY, student_name VARCHAR(50), gender VARCHAR(10), city VARCHAR(50), join_year INT );
CREATE TABLE courses ( course_id INT PRIMARY KEY, course_name VARCHAR(100), department VARCHAR(50) );
CREATE TABLE marks ( mark_id INT PRIMARY KEY, student_id INT, course_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id) );
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
```

student_id	student_name	gender	city	join_year
1	Anu	F	Tumakuru	2024
2	Ravi	M	Bengaluru	2023
3	Kiran	M	Tumakuru	2024
4	Sneha	F	Mysuru	2023
5	Manu	M	Tumakuru	2022

2. Display only student_name and city from students table.

```
create database college;
use college;
CREATE TABLE students ( student_id INT PRIMARY KEY, student_name VARCHAR(50), gender VARCHAR(10), city VARCHAR(50), join_year INT );
CREATE TABLE courses ( course_id INT PRIMARY KEY, course_name VARCHAR(100), department VARCHAR(50) );
CREATE TABLE marks ( mark_id INT PRIMARY KEY, student_id INT, course_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id) );
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
```

student_name	city
Anu	Tumakuru
Ravi	Bengaluru
Kiran	Tumakuru
Sneha	Mysuru
Manu	Tumakuru

3. Show all courses.

```
create database college;
use college;
CREATE TABLE students ( student_id INT PRIMARY KEY, student_name VARCHAR(50), gender VARCHAR(10), city VARCHAR(50), join_year INT );
CREATE TABLE courses ( course_id INT PRIMARY KEY, course_name VARCHAR(100), department VARCHAR(50) );
CREATE TABLE marks ( mark_id INT PRIMARY KEY, student_id INT, course_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id) );
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
```

course_id	course_name	department
101	SQL Basics	Computer Science
102	Excel for Analysts	Commerce
103	Statistics	Mathematics

4. Display students who are from Tumakuru.

```
create database college;
use college;
CREATE TABLE students ( student_id INT PRIMARY KEY, student_name VARCHAR(50), gender VARCHAR(10), city VARCHAR(50), join_year INT );
CREATE TABLE courses ( course_id INT PRIMARY KEY, course_name VARCHAR(100), department VARCHAR(50) );
CREATE TABLE marks ( mark_id INT PRIMARY KEY, student_id INT, course_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id) );
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
```

student_id	student_name	gender	city	join_year
1	Anu	F	Tumakuru	2024
3	Kiran	M	Tumakuru	2024
5	Manu	M	Tumakuru	2022

5. Display students who joined in 2024.

```
create database college;
use college;
CREATE TABLE students ( student_id INT PRIMARY KEY, student_name VARCHAR(50), gender VARCHAR(10), city VARCHAR(50), join_year INT );
CREATE TABLE courses ( course_id INT PRIMARY KEY, course_name VARCHAR(100), department VARCHAR(50) );
CREATE TABLE marks ( mark_id INT PRIMARY KEY, student_id INT, course_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id) );
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
select * from students where join_year=2024;
```

student_id	student_name	gender	city	join_year
1	Anu	F	Tumakuru	2024
3	Kiran	M	Tumakuru	2024

6. Show students whose gender is F.

```
CREATE TABLE students ( student_id INT PRIMARY KEY, student_name VARCHAR(50), gender VARCHAR(10), city VARCHAR(50), join_year INT );
CREATE TABLE courses ( course_id INT PRIMARY KEY, course_name VARCHAR(100), department VARCHAR(50) );
CREATE TABLE marks ( mark_id INT PRIMARY KEY, student_id INT, course_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id) );
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
select * from students where join_year=2024;
select * from students where gender='F';
```

student_id	student_name	gender	city	join_year
1	Anu	F	Tumakuru	2024
4	Sneha	F	Mysuru	2023

7. Show marks greater than 80.

```
CREATE TABLE students ( student_id INT PRIMARY KEY, student_name VARCHAR(50), gender VARCHAR(10), city VARCHAR(50), join_year INT );
CREATE TABLE courses ( course_id INT PRIMARY KEY, course_name VARCHAR(100), department VARCHAR(50) );
CREATE TABLE marks ( mark_id INT PRIMARY KEY, student_id INT, course_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id) );
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
select * from students where join_year=2024;
select * from students where gender='F';
select marks from marks where marks>80;
```

marks
85
90
88
81

8. Display course names from Commerce department.

```
CREATE TABLE marks ( mark_id INT PRIMARY KEY, student_id INT, course_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id) );
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
select * from students where join_year=2024;
select * from students where gender='F';
select marks from marks where marks>80;
select course_name from courses where department='commerce';
```

course_name
Excel for Analysts

9. Show students who are not from Bengaluru.

```
FOREIGN KEY (course_id) REFERENCES courses(course_id) );
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
select * from students where join_year=2024;
select * from students where gender='F';
select marks from marks where marks>80;
select course_name from courses where department='commerce';
select * from students where not city='bengaluru';
```

student_id	student_name	gender	city	join_year
1	Anu	F	Tumakuru	2024
3	Kiran	M	Tumakuru	2024
4	Sneha	F	Mysuru	2023
5	Manu	M	Tumakuru	2022

10. Display marks between 70 and 90.

```
FOREIGN KEY (course_id) REFERENCES courses(course_id) );
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
select * from students where join_year=2024;
select * from students where gender='F';
select marks from marks where marks>80;
select course_name from courses where department='commerce';
select * from students where not city='bengaluru';
select marks from marks where marks between 70 and 90;
```

marks
85
72
90
88
79
81

11. Display all students ordered by student_name ascending.

```
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024), (2, 'Ravi', 'M', 'Bengaluru', 2023), (3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
select * from students where join_year=2024;
select * from students where gender='F';
select marks from marks where marks>80;
select course_name from courses where department='commerce';
select * from students where not city='bengaluru';
select marks from marks where marks between 70 and 90;
select * from students order by student_name;
```

student_id	student_name	gender	city	join_year
1	Anu	F	Tumakuru	2024
3	Kiran	M	Tumakuru	2024
5	Manu	M	Tumakuru	2022
2	Ravi	M	Bengaluru	2023
4	Sneha	F	Mysuru	2023

12. Show marks ordered from highest to lowest.

```
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
select * from students where join_year=2024;
select * from students where gender='F';
select marks from marks where marks>80;
select course_name from courses where department='commerce';
select * from students where not city='bengaluru';
select marks from marks where marks between 70 and 90;
select * from students order by student_name;
select marks from marks order by marks desc;
```

marks
90
88
85
81
79
72

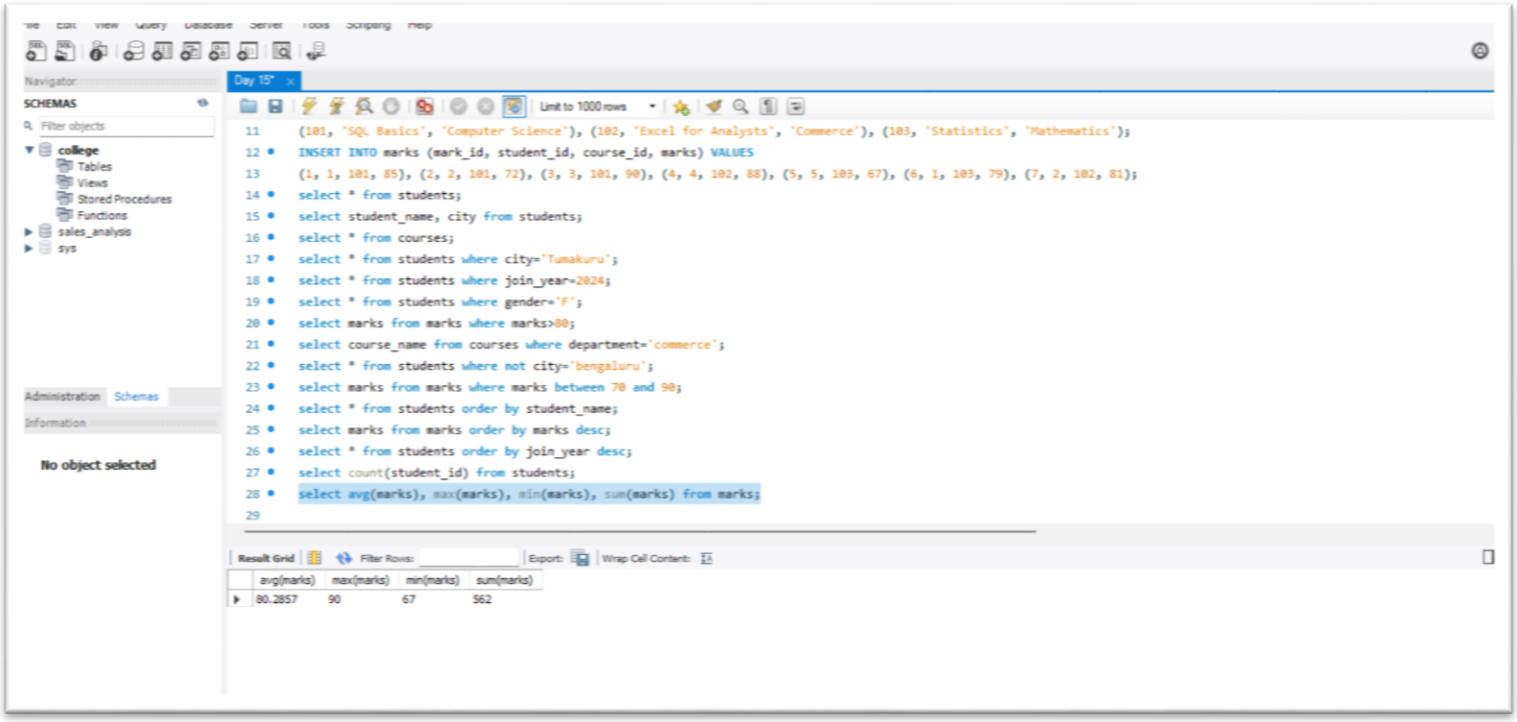
13. Display students ordered by join_year descending.

```
(4, 'Sneha', 'F', 'Mysuru', 2023), (5, 'Manu', 'M', 'Tumakuru', 2022);
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
select * from students where join_year=2024;
select * from students where gender='F';
select marks from marks where marks>80;
select course_name from courses where department='commerce';
select * from students where not city='bengaluru';
select marks from marks where marks between 70 and 90;
select * from students order by student_name;
select marks from marks order by marks desc;
select * from students order by join_year desc;
```

student_id	student_name	gender	city	join_year
1	Anu	F	Tumakuru	2024
3	Kiran	M	Tumakuru	2024
2	Ravi	M	Bengaluru	2023
4	Sneha	F	Mysuru	2023
5	Manu	M	Tumakuru	2022

14. Find total number of students.

```
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'), (102, 'Excel for Analysts', 'Commerce'), (103, 'Statistics', 'Mathematics');
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90), (4, 4, 102, 88), (5, 5, 103, 67), (6, 1, 103, 79), (7, 2, 102, 81);
select * from students;
select student_name, city from students;
select * from courses;
select * from students where city='Tumakuru';
select * from students where join_year=2024;
select * from students where gender='F';
select marks from marks where marks>80;
select course_name from courses where department='commerce';
select * from students where not city='bengaluru';
select marks from marks where marks between 70 and 90;
select * from students order by student_name;
select marks from marks order by marks desc;
select * from students order by join_year desc;
select count(student_id) from students;
```

count(student_id)
5

15. Find average marks.
16. Find highest marks.
17. Find lowest marks.
18. Find total marks scored by all students.